Where should the training data of a neural chemistry surrogate come from? Online learning from direct integration versus physics-based re-exploration seeded by the simulation

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Abstract

Neural-network surrogates of finite-rate chemistry trained a priori on canonical reactors meet, in the actual simulation, states they have not seen. Two ways of closing that gap with the help of the simulation are compared on the same laminar methane/air burner, the same 18-species mechanism (GA18), the same network architecture and the same evaluation protocol against direct integration (DI). In *ISANN* (in situ adaptive neural network), the running simulation trains its own surrogate: it starts with DI, stores deduplicated (state, time step, DI result) pairs, fine-tunes a pre-trained network on them and switches it in through an acceptance gate. In *ITERML* (iterative machine learning), no state of the simulation is ever a training pair: a run with the network collects the states where the network is out of distribution or disagrees with sparse DI checks, these states are deduplicated and used only as starting points of new Cantera reactor explorations (partially stirred reactors, relaxation, ignition, cooling), and the network is retrained on physics data alone. Run without any DI fallback (the only mode available in commercial solvers), the final ISANN network reached a mean flame-zone

temperature error of 2.1 K over 25–50 ms, and the first ITERML iteration 3.3 K (1.7 K at 50 ms), against 68.8 K for the a priori network. Their start-up errors are mirror images: the ISANN network ignites the fuel jet 0.4 ms early, the ITERML network 0.6 ms late. ISANN gives better radicals and CO; ITERML keeps the training data independent of the simulation, so the model remains a physics-based model of the mechanism, at the cost of a slower growth of CO and heat release in the hot gas and a small loss on an offline benchmark of canonical states. A second ITERML iteration, aimed at the late ignition, made the network worse (6.6 K, ignition 1.1 ms late, total CO 6.4 times DI): its errors build up over many steps in cooled lean gas whose CO level lies below the absolute floor of the one-step check, so the check never saw them. Physics-only seeding works, but only as well as the error detector that chooses the seeds.

Keywords: Chemistry acceleration, Neural network surrogate, Online learning, Active learning, Thermochemical space exploration, Out-of-distribution detection

1. Introduction

The integration of stiff finite-rate chemistry dominates the cost of many reacting-flow simulations. Storage/retrieval methods such as in situ adaptive tabulation (ISAT) [1] tabulate direct integrations during the simulation itself; regression methods replace the integrator with a fitted map, and artificial neural networks (ANN) have been used for this since the 1990s [2, 3]. The training data of such networks have come from the simulation of a similar case [4], from canonical laminar flames [5], from abstract problems that sample the composition space [6, 7, 8], or have been used to fit neural ordinary

differential equations [9].

Whatever the source, an a priori data set is a guess of the states the simulation will visit, and the guess is never complete. Out-of-distribution (OOD) detectors with a DI fallback limit the damage, but a fallback does not exist in every solver (a commercial code such as Ansys Fluent calls a user-defined chemistry step that must return a result), and errors that build up over many steps inside the training data are not detected by an OOD test. The simulation itself is the only reliable source of information on the states it needs. The question this paper addresses is how that information should enter the surrogate. Two answers are possible:

1. *as labels*: the simulation integrates the states it visits with DI and the network is trained on those pairs. This is the ISAT idea transferred to a network; we call the scheme ISANN;
2. *as locations*: the simulation only says *where* the network fails, and the training data are generated by physics-based exploration started from those states. The data stay a model of the mechanism, not of the simulation; we call the loop ITERML.

The first answer uses the most relevant data possible but ties the network to one simulation, and its labels are produced by a solution that the network may already have biased. The second keeps the training data independent of the CFD solution, which matters when the surrogate is to be reported as a physics-based model, reused in another configuration, or validated against the same simulation it was trained for, but it relies on the exploration to reproduce the relevant states from a seed.

We compare the two on the same burner, mechanism, network, explo-

ration framework, OOD detector and evaluation protocol, and report accuracy against DI during start-up and in the established flame, species and heat release, generality on an offline benchmark of canonical states, and cost. [TODO: Literature on online/active learning of chemistry surrogates and adaptive sampling (e.g. active learning of reaction manifolds, on-the-fly ANN tabulation, DeepFlame and similar frameworks); position both schemes against it.]

2. Common elements

2.1. Surrogate network

The surrogate maps the composition at the start of a chemistry step to that at its end, at constant pressure and enthalpy. Species mass fractions are Box–Cox transformed ($\lambda = 0.1$) and standardised; the context $c = (T, p, h, \log_{10} \Delta t)$ modulates a residual MLP through feature-wise linear modulation (FiLM) [10] (three residual blocks, 192 hidden units, SiLU, layer normalisation, output initialised to zero so that the untrained network is the identity). The temperature at the end of the step follows from the predicted composition and the conserved enthalpy. Steps longer than the trained range ($\Delta t_{\max} = 4 \times 10^{-5}$ s) are sub-stepped; cells below $T_{\text{freeze}} = 600$ K are not advanced. The same C core is called by an OpenFOAM v2412 chemistry solver and by an Ansys Fluent DEFINE_CHEM_STEP user-defined function.

2.2. A priori exploration

The a priori data come only from Cantera [11] calculations with the GA18 mechanism at the burner's inlet conditions: 0D ignition and equilibrium

relaxation, partially stirred reactors (PaSR), steady and unsteady flamelets over a range of strain rates, vitiated mixing and heat-loss (cooling) reactors (regimes 0–8). The pairs are integrated over a range of chemistry steps taken from the histogram of the per-cell chemistry time step of the target solver. Successive revisions of this exploration produced the networks ood1 to ood8; ISANN starts from ood7 and ITERML from ood8, the best a priori network at the time each scheme was started (Section 4.1). [TODO: Table of the a priori regimes and pair counts of ood7 and ood8.]

2.3. Out-of-distribution detection and shadow checks

Each exported model carries tables built from its own training data. A state is OOD if its temperature, a species or the pressure leaves the trained range (globally or in its mixture-fraction band), if its residual to a local principal-component manifold is too large, or if it is farther than d_{L} from the nearest of a set of training landmarks. *Shadow checks* integrate a sample of cells with DI from the same state as the network; the sample is stratified over bins of (temperature, mixture fraction, inside/OOD) so that it covers all regions of the flame and not mostly the far field. A check fails if $|T_{\mathrm{ANN}} - T_{\mathrm{DI}}| > \max(1\,\mathrm{K},\, 0.3|T_{\mathrm{DI}} - T_0|)$ or if the analogous test on OH (and in ITERML also H, O and CO; floor 5×10^{-5}) fails.

2.4. Deduplication

Both schemes store states with the same two-level deduplication: a grid in T, $\sqrt{Z}$, the logarithm of eight key species and $\log_{10} \Delta t$ with at most one state per box (reservoir sampling), and a radius of novelty in the standardised coordinates of the OOD detector. On 902 029 hot-cell DI states of the burner

the two levels reduce the states 93 times while every state stays within $0.27\,d_{\mathrm{L}}$ of a kept one. ITERML adds a third level: a seed within the radius of any seed of an earlier iteration is dropped.

2.5. Test case

The burner is a laminar, non-premixed methane/air burner of a domestic cooking geometry, computed 2D axisymmetric (5° wedge, 60k cells) with gravity. Pure CH_4 enters at 300 K through a central tube ($r < 2.5$ mm, $6.7{\times}10^{-6}$ kg/s, about 340 W), air through an annular co-flow and a slow outer co-flow; a pot with convective walls ($h = 15$ W m^{-2}K^{-1}, 323 K) sits above the burner, and the only open boundary is the lateral side. The mechanism is the 18-species GA18 skeletal mechanism [TODO: reference]. The flow solver is rhoReactingBuoyantFoam (laminar, Euler implicit, $\Delta t = 2 \times 10^{-5}$ s). The simulation starts from a lit state (lean burnt gas at 1480 K under the pot, air elsewhere); the reference is a DI run from the same state. [TODO: Geometry figure and DI temperature field at 50 ms.]

2.6. Evaluation protocol

Every model is scored with one fixed protocol (v1): the lit start above, snapshots every 0.5 ms up to 5 ms and every 2.5 ms afterwards, all compared cell by cell with the DI run. The *start-up* phase (0–5 ms) is scored by the ignition time (first snapshot with $T_{\max} > 1600$ K) and the temperature error at the nozzle; the *established flame* (25–50 ms here) by the mean $|T - T_{\mathrm{DI}}|$ in the flame zone ($T > 1600$ K in either solution; headline metric), its drift, the error by zone, the L1 error and domain total of OH, CO, CH_4 and H in hot cells ($T > 600$ K), the volume above 1500 K, and the heat-release rate

obtained by evaluating GA18 on each state. Candidate models are run in *pure* mode, without any DI fallback, because this is the only mode the target Fluent case offers; a run is valid only if it reaches 50 ms. The CFD states are diagnostic data of the protocol and are never used for training in ITERML.

2.7. Offline benchmark

To measure generality away from the burner, each model is also run in Cantera on a fixed set of canonical states that contain no CFD state (equilibrium, cooled, lean hot, vitiated, partially burnt, flamelet), 2500 steps of 2×10^{-5} s in homogeneous constant-(p, h) reactors. The score is the mean rank over one-step increment errors, radical errors at 1 ms, fixed-point errors and temperature errors at 10 ms and 50 ms (lower is better).

3. ISANN: learning from the run's direct integration

The run starts with DI in every cell (*DI phase*); every hot cell offers its (state, Δt, DI result) pair to the store. The first network is fine-tuned from the parent (ood7) when at least 100 steps have run, 3 000 deduplicated pairs exist and the store has saturated (or at 400 steps). Afterwards, cells go to DI only if they are OOD or belong to an *untrusted bin* (a bin whose decayed shadow failure count exceeds its successes), and a new round starts after 100 steps if 1 500 new pairs exist, the DI share exceeds 5%, or a bin is untrusted. A round fine-tunes the current network (20 epochs, learning rate 2×10^{-4}) on the online pairs plus a threefold replay of a priori pairs, with 8-step Cantera continuations of online states in the rollout loss. A candidate is published only if its one-step error on held-out online pairs is lower than the current network's and its 100-step 0D rollouts do not fail more often nor have a

median error above 1.1 times the current one. The trainer runs beside the CFD on two of the four cores and publishes models through a control file read by all ranks.

4. ITERML: the simulation chooses where to explore

Figure 1 compares the two loops. One ITERML iteration has six stages:

A *Collection run.* The burner is run in hybrid mode with the current network (OOD cells to DI so that the visited states stay realistic, and without the induction-zone DI patch so that the network's own ignition errors appear). The store keeps, with the grid cap, every state that is OOD, fails a shadow check, belongs to a bin with more than 20% shadow failures, or starts a drifting *sentinel*: induction-zone cells (900–1700 K) copied and integrated for 25 steps with both the network and DI, a drift above 50 K or a factor 2 in OH flagging the start state. The run stops when the store is saturated or full.

B *Seeds.* The states are merged, deduplicated (three levels), described by their flame features (Bilger Z, heat-loss fraction ψ, progress, radical excess) and assigned by simple rules to a seeded regime: 10 PaSR with IEM mixing against the inlet streams, 11 relaxation of burnt gas with radical perturbations, 12 ignition, 13 cooling.

C *Seeded exploration.* Each seed spawns jittered Cantera reactors of its regime (element-conserving radical perturbations, $\psi \pm 0.05$, small mixing with either stream). The seed state itself is never paired, and any pair that starts within 1×10^{-8} of a seed is dropped, so not even a numerical copy of a CFD state reaches the training set. Seeded pairs

Table 1: Design differences between the two schemes.

	ISANN	ITERML
Role of the CFD states	training pairs (state, Δt, DI result)	seeds: where to explore
Origin of the training labels	DI of the running CFD	Cantera reactors started from seeds
CFD state in the training set	yes	never (excluded, with a 1×10^{-8} margin)
When the network changes	during the run (control file)	between runs
DI needed in the production run	yes (DI phase, fallback)	no (pure mode)
Error detection	OOD + shadow checks	OOD + shadow checks + multi-step sentinel
Parent network	ood7	ood8
Model label	adapted to this simulation	a priori + seeded exploration

are capped at 35% of the training set; from iteration 2 the newest iteration receives at least half of that budget and earlier seeded pairs are thinned.

D *Training.* Warm start from the previous network on the a priori plus all seeded pairs.

E *Gate.* The offline benchmark; the candidate may not be worse than its parent by more than the spread between training seeds (15 K at 10 ms).

F *Evaluation.* Protocol v1 in pure mode, 0–50 ms. The loop stops when the flame error is below 5 K with a drift below 2 K per 10 ms, ignition within 0.3 ms of DI, total OH within $\times$0.7–1.4 and less than 1% OOD cells, or when two iterations improve by less than 10%.

Table 1 summarises the differences.

4.1. Fairness of the comparison

The comparison is not perfectly controlled, and the differences are stated here rather than hidden in the results. (i) The parents differ (ood7 for

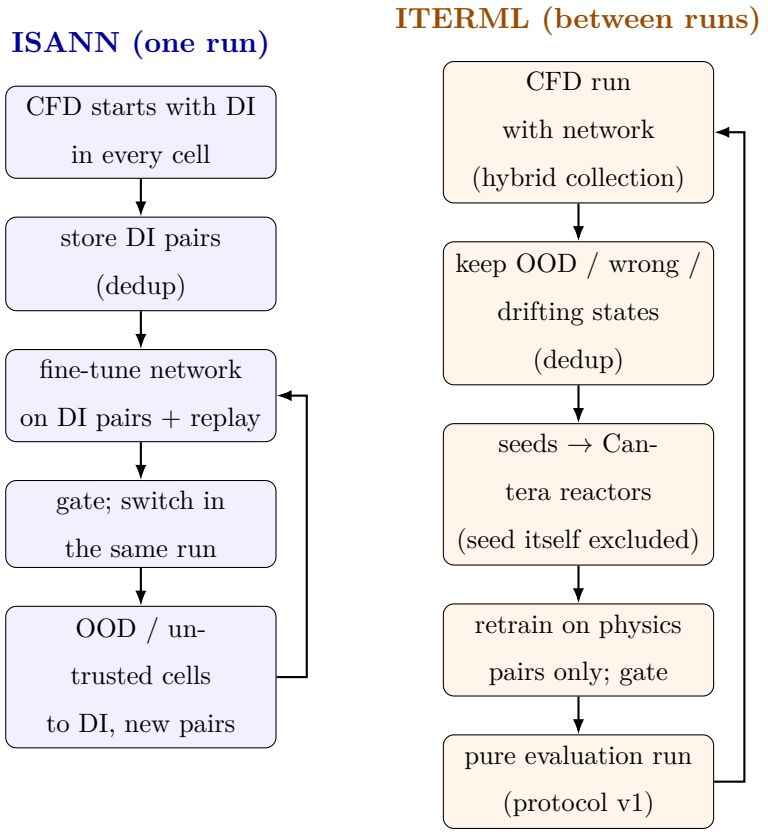


Figure 1: The two loops. In ISANN the CFD's DI results are the training labels and the network changes during the run. In ITERML the CFD only provides starting points for physics-based exploration; training happens between runs and on Cantera data alone.

ISANN, ood8 for ITERML); ood8 has a better a priori coverage of the flame but, like ood7, ignites the fuel jet early in pure mode. (ii) ISANN's headline run is hybrid, by design; the fair counterpart of an ITERML evaluation is therefore the final ISANN network (v6) run alone in pure mode under the same protocol, which we report as the main ISANN result. (iii) The pure v6 run used two ranks and the ITERML evaluation four, so costs per step are not directly comparable. (iv) Each network was trained once; the spread

between training seeds on the offline benchmark is about 15 K at 10 ms, comparable with some of the differences reported. [TODO: Repeat ISANN from ood8, or ITERML from ood7, to remove (i); retrain both finals with three seeds.]

5. Results

5.1. How much information each scheme takes from the CFD

ISANN's first run kept its DI phase to 11.2 ms, accepted six networks (v1–v6, at 11.2, 15.4, 20.0, 23.7, 28.7 and 36.2 ms) in twelve rounds, and trained on about 35 000 deduplicated DI pairs of the run, replayed with three a priori pairs each; about a quarter of every training batch was therefore CFD data.

ITERML's first collection run (iteration-0 network ood8) stopped at 9.2 ms (458 steps) with 16 033 stored states. Of these, 13 732 were seed candidates and 7 891 remained after the radius of novelty (Table 3); 70% are lean ($Z < Z_{\mathrm{st}} = 0.055$). The 2372 shadow checks of the run failed in 4.3% of the cases, and one of 31 sentinels drifted. The seeded exploration produced 120 719 pairs (the 35% cap) in 7 min, and none of them is a CFD state. The second collection run, with the iteration-1 network, a sentinel four times as dense and a wider ignition rule (fuel and O_2 at 800–1700 K), stored about 15 200 states, which gave 2 912 seeds after deduplication against iteration 1, 173 of them ignition seeds (Table 2). Only 1.5% of its 8211 shadow checks failed and none of its 720 sentinels drifted. The iteration-2 seeds produced 60 359 seeded pairs, and the iteration-1 seeded pairs were thinned to the same number so that together they stayed within the 35% cap. Warm-start training from iteration 1 took 33 min, and the network passed the offline gate

Table 2: ITERML seeds of iteration 2 (iteration-1 network, denser sentinel, wider ignition rule), deduplicated against the seeds of iteration 1.

Regime	Seeds	Weight	T median [K]	Z median	ψ median
10 PaSR	576	804	642	0.007	−0.09
11 relaxation	1901	4824	1789	0.051	0.15
12 ignition	173	305	875	0.093	0.64
13 cooling	262	475	696	0.108	0.74

Caught as: untrusted bin 1424; OOD 1040; untrusted bin + shadow error 354; shadow error 93; OOD + shadow error 1; sentinel drift 0.

with a clear margin (Section 5.5).

The single ignition seed of iteration 1 is the main lesson of this stage: ignition errors build over many steps in cells where every single step is inside the training data and passes a one-step shadow check, so neither OOD flags nor shadow checks catch them, and the first sentinel was too sparse. ISANN does not have this problem during its own run, because its DI phase integrates the ignition exactly; it shows it when the trained network is reused from the start (Section 5.3).

5.2. Established flame

Figure 2 and Table 4 show the errors against DI. In the established flame (25–50 ms) all the adapted networks are within a few kelvin of DI, while the a priori ood1 network in pure mode is 68.8 K off and drifts. The ISANN v6 network alone gives 2.1 K with a bias of +0.8 K and a drift of −0.5 K per 10 ms; ITERML iteration 1 gives 3.3 K with a bias of −1.5 K. The ITERML error is still falling at the end of the window (from 5.3 K at 25 ms to 1.7 K at

Table 3: ITERML seeds of iteration 1, by seeded regime and by the reason they were caught. Weight: CFD states merged into the seeds of the regime.

Regime	Seeds	Weight	T median [K]	Z median	ψ median
10 PaSR	2401	3791	1029	0.025	0.16
11 relaxation	4924	9075	1638	0.047	0.12
12 ignition	1	1	1508	0.129	0.04
13 cooling	565	865	749	0.028	0.63
Caught as: OOD 3104; untrusted bin + shadow error 2400; untrusted bin 2263; shadow error 93; OOD + shadow error 30; sentinel drift 1.					

47.5–50 ms, a drift of −1.6 K per 10 ms), so its 50 ms value equals that of v6. By zone, ITERML's largest errors are in the reactive (5.5 K) and rich (7.4 K) zones, against 1.8 and 3.9 K for v6; the 95th percentile of the hot-cell error is 5 K against 2 K. The ISANN run itself, with DI fallback and changing networks, gives 2.0–3.4 K (mean 2.6 K); it is not lower than v6 alone because the network changed during the window. Both pure networks reproduce the volume above 1500 K within 1% and the position of the flame tip and base on the axis to better than the 0.5 mm resolution of the analysis. The ITERML evaluation run had 0.9% of the hot cells OOD during the flame phase (none of them sent to DI), and its 4571 shadow checks had a median one-step error of 0.027 K and a 5% error of the temperature increment.

ITERML iteration 2 was worse than iteration 1 in the flame: 6.6 K over 25–50 ms with a bias of −4.1 K, falling from 8.4 K at 25 ms to 5.0 K at 50 ms (drift −1.3 K per 10 ms), so it misses the 5 K target that iteration 1 met. Its zone errors are similar to those of iteration 1 (reactive 5.4 K, rich

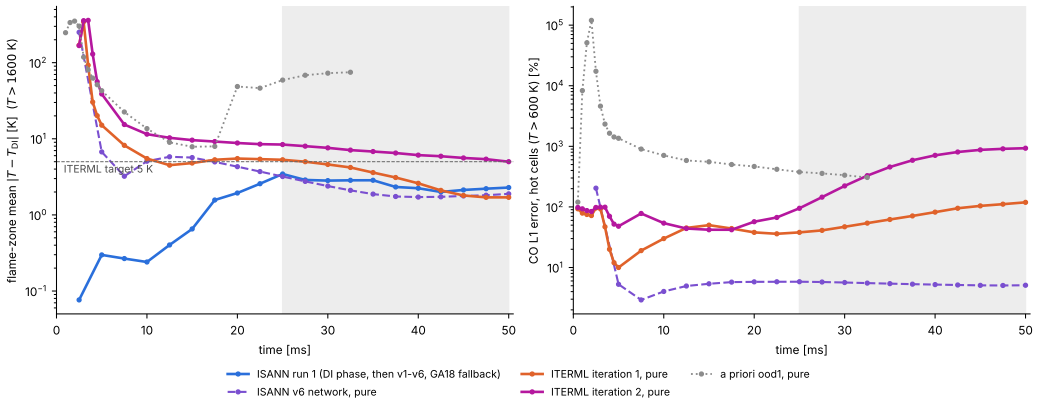


Figure 2: Left: mean temperature error in the flame zone ($T > 1600$ K) against DI. Right: L1 error of CO in hot cells ($T > 600$ K). Shaded: the scored flame window. ISANN run 1 has the DI phase up to 11.2 ms, so its early error is that of a restart; the pure runs use the network in every cell from the first step. The pure v6 run was written every 2.5 ms only. ood1 pure was stopped at 32.5 ms.

5.8 K, hot-cell 95th percentile 6 K), so the larger flame error is a systematic cold bias rather than a few bad cells; its OOD share was 0.7%. The one-step shadow checks of the evaluation run were as good as those of iteration 1 (median 0.04 K, 6% error of the temperature increment): one-step accuracy did not predict the outcome.

5.3. Start-up and ignition

The start-up separates the two schemes most clearly (Fig. 3). Every a priori network of this project ignites the fuel jet early in pure mode: ood1 by 1.6 ms, ood8 by about 1.5 ms (the iteration-0 network of ITERML; +348 K at 2 ms in the ITERML test run). The ISANN network v6, adapted with DI pairs that include the ignition, still ignites 0.4 ms early (+251 K in the first flame cells at 2.5 ms): its training pairs come from a DI phase in which the

Table 4: Protocol v1 against GA18 DI. Pure: network in every cell, no DI fallback. Flame: mean $|T - T_{\mathrm{DI}}|$ for $T > 1600$ K over 25–50 ms. Totals: domain amount, model/DI, mean over 25–50 ms. DI ignites at about 2.6 ms.

	ISANN run 1 (hybrid)	ISANN v6 (pure)	ITERML iter. 1 (pure)	ITERML iter. 2 (pure)	ood1 (pure)
Training data from the CFD	DI pairs	DI pairs	none	none	none
Ignition time, model − DI [ms]	0 (DI phase)	−0.4	+0.6	+1.1	−1.6
Worst flame bias at start-up [K]	–	+251	−345	−360	+351
Flame $\lvert dT\rvert$, 25–50 ms [K]	2.6	2.1	3.3	6.6	68.8[a]
Flame bias [K]	−0.6	+0.8	−1.5	−4.1	−51[b]
Drift [K per 10 ms]	–	−0.5	−1.6	−1.3	+20[a]
Flame $\lvert dT\rvert$ at 50 ms [K]	2.3	1.9	1.7	5.0	–
OH total, model/DI	–	0.55	1.05	2.15	–
CO total, model/DI	–	0.97	1.66	6.38	–
CH_4 total, model/DI	–	1.00	1.27	0.99	–
H total, model/DI	–	0.89	1.19	1.19	–
CO L1, hot cells	–	5%	75%	551%	300–400%
Heat release, model/DI	–	0.92[c]	1.21	1.10	–
GA18 / OOD share, flame phase	0.14–0.5% / <0.2%	0 / 0	0 / 0.9%	0 / 0.7%	0 / –
Cost [s per step]	4.2 (2 ranks)	2.85 (2 ranks)	1.99 (4 ranks)	1.91 (4 ranks)	–

[a] 25–32.5 ms, run stopped at 32.5 ms. [b] at 25 ms. [c] flame zone only.

ignition happened once, and the deduplication keeps one pair per region, so the few induction pairs weigh little. ITERML iteration 1 removed the early ignition but overshot: the jet ignites 0.6 ms late (−345 K at 3.0 ms), after which the error decays to 15 K at 5 ms. The seeded relaxation and PaSR data (Table 3) slowed the radical build-up in the induction zone, and the single ignition seed could not balance it. Iteration 2 was designed for exactly this: a four-times denser sentinel and an ignition rule that sends every seed with fuel and oxygen at 800–1700 K to the ignition reactors, giving 173 ignition seeds.

It did not work. The iteration-2 network ignites the jet at 3.6 ms, 1.1 ms

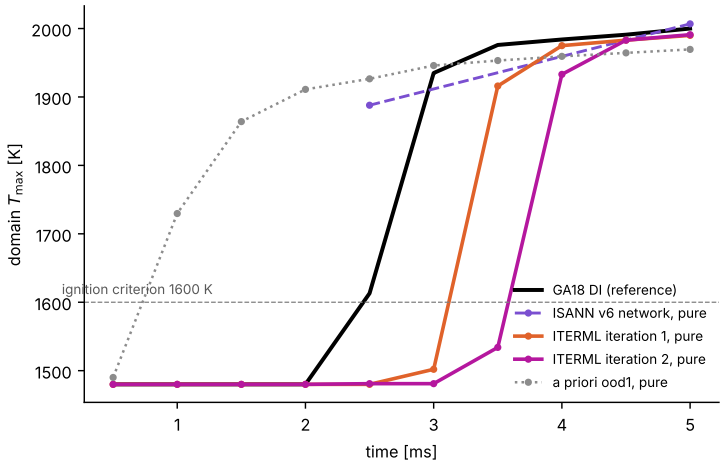


Figure 3: Maximum temperature in the domain during start-up, pure runs, against DI. The dashed line is the ignition criterion of the protocol. The pure v6 run has no snapshot before 2.5 ms.

after DI, and the flame is still 360 K too cold at 3.5 ms (Fig. 3). The 173 ignition seeds were caught mostly by the wider rule (median 875 K, rich, strongly cooled, $\psi = 0.64$), not by the sentinel, which flagged none of its 720 tracks: they describe cold, heat-losing mixtures near the walls rather than the induction zone at the base of the jet, and their ignition reactors added slow chemistry where the jet needed fast chemistry. A seed rule based on temperature and composition alone does not identify the cells that set the ignition delay; the sentinel does, but only if it is placed in the induction zone and run long enough to drift.

5.4. Species and heat release

Away from temperature the ISANN network is clearly better. With v6, total CO is within 3% of DI and its hot-cell L1 error is 5%; total OH is 0.55 of DI. With ITERML iteration 1, total OH is right (1.05), but CO, CH_4 and

H are 27–66% too high and the CO error grows steadily in the flame window, from 38% at 25 ms to 119% at 50 ms (Fig. 2, right), together with the heat release evaluated on the states (from 1.04 to 1.43 times DI, Fig. 4). The error is not in the flame, where the CO L1 error is 2%, but in cooled lean burnt gas (400–1000 K, $Z \approx 0.027$) near the walls and the hood, where at 50 ms CO is about ten times that of DI and the gas is 15–32 K too cold. In that gas the remaining CO oxidises slowly over residence times far longer than the 20 ms of the seeded relaxation and cooling reactors. ISANN does not have this problem because its DI pairs are taken in that very gas.

In iteration 2 this error grew: total CO is 6.4 times DI, its hot-cell L1 error rises from 95% at 25 ms to 931% at 50 ms, and CO in the cooled lean gas reaches about 100 times DI at 50 ms; total OH is 2.15 times DI. The flame-zone CO is still right (L1 3%, total in the flame 0.97 of DI). The shadow check could not see this error: its CO test has an absolute floor of 5×10^{-5}, above the CO mass fractions of that gas, so every one-step error there passes, while the many small steps pile up exactly as in the ignition delay. The seeds that would have targeted it were therefore never caught, and the iteration-2 data, with more relaxation and fewer cooling seeds, moved that gas further from DI. A relative CO and OH test (floor 1×10^{-7}) and sentinels in cooled lean gas are the obvious corrections.

5.5. Generality away from the burner

On the offline benchmark the two schemes behave differently. The ISANN v6 network ranked first among the networks of the project on 440 canonical states (score 1.75 against 2.38 for its parent ood7 and 2.88 for ood1): fine-tuning on the flame's DI pairs with replay did not cost generality. The

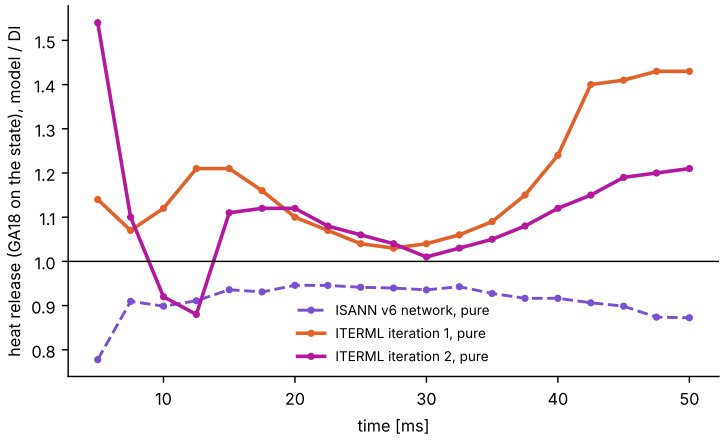


Figure 4: Heat-release rate obtained by evaluating GA18 on the model's state, over the domain, relative to DI, pure runs.

ITERML iteration-1 network lost some accuracy on the 218-state set against its parent ood8 (mean $|dT|$ at 10 ms 31.8 against 28.2 K, at 50 ms 59 against 51 K; equilibrium fixed point 42 against 27 K), within the seed-spread tolerance of the gate. The loss is concentrated in vitiated states (85 against 53 K at 10 ms), while partially burnt and cooled states improved (69 against 103 K at 10 ms, and 11 against 34 K at 50 ms). A 35% share of seeded data drawn from one burner's burnt gas thus displaces part of the a priori coverage, even though no CFD state is in it. Iteration 2 shows that the benchmark does not predict the CFD either: it ranked the iteration-2 network clearly better than iteration 1 (mean $|dT|$ 22.9 against 31.8 K at 10 ms and 40 against 59 K at 50 ms, mostly through the vitiated family, 34 against 85 K), yet it was the worse network in the burner. A gate made only of canonical states cannot see errors that build up in the burner's slow, cold, long-residence gas. [TODO: Run both final networks on the same 440-state set with three training seeds

each; the two sets above are not the same.]

5.6. Cost

ISANN pays for its labels during the production run: the DI phase cost 7.9 s per step on two ranks and the network phase 4.2 s, and the first 50 ms took 3.5 h; once trained, v6 alone ran at 2.85 s per step on two ranks. ITERML pays between runs: iteration 1 took about 2.5 h of wall time on four cores (collection run 18 min, seeds and seeded exploration 7 min, warm-start training 32 min, gate 2 min, pure evaluation to 50 ms the rest), and the pure network ran at 1.99 s per step on four ranks. The production run of an ITERML network needs no DI at all, so it can run in a solver without a hybrid mode; a reused ISANN network (Section 5.3) has the same property but carries CFD data in its training set. [TODO: Same-rank timing of v6 and ITERML networks; DI cost on four ranks.]

6. Discussion

Labels or locations. Using the simulation's DI results as labels gives the most accurate network for that simulation: v6 is the best network of the project on flame temperature, CO and heat release. Using the simulation only to choose locations reaches the same flame temperature by 50 ms with training data that remain a model of the mechanism, but it converges more slowly and depends on the exploration reproducing the relevant states from a seed. Where the physics of the seeded reactor is right (burnt-gas relaxation, PaSR mixing), one iteration is enough; where it is not (long-residence CO oxidation, ignition from a handful of seeds), the error is visible in the next evaluation

and points to the regime that needs work, which is a diagnostic ISANN does not offer.

What each detector sees. Both schemes rely on OOD flags and one-step shadow checks, and both miss multi-step errors inside the training data: ISANN's v6 network ignites early without a single OOD cell or failed check, and ITERML's first collection caught one ignition state out of 16 033. The multi-step sentinel of ITERML is the only element of either scheme that targets this error, and its density matters.

Mirror-image ignition. The a priori and ISANN networks ignite early, ITERML's first network late. The ignition delay of the jet is set by a few cells in the induction zone; a network that is slightly too reactive there ignites early and one slightly too slow ignites late, and neither shows up in the flame-zone error once the flame is established. The ignition time must therefore be scored separately, as protocol v1 does, and should be part of the acceptance gate of both schemes.

When to use which. For repeated simulations of one configuration where DI can run in the production solver, ISANN gives the best accuracy at the lowest effort. When the surrogate must run in a solver without DI fallback, must be reported as a physics-based model, or will be transferred to other conditions, ITERML keeps the training data clean and still uses the simulation to direct the effort. After two iterations ITERML has not reached ISANN's accuracy, and the second iteration shows why: the loop only improves the network where its detectors look.

The detector sets the ceiling. In ITERML the CFD contributes nothing but the choice of seeds, so the error detector is the whole link between the simulation and the training data. Both failures of iteration 2 were errors that are small per step and large over many steps, in regions where the one-step check is blind (ignition delay at the jet base, CO below an absolute floor in cooled gas). ISANN is much less sensitive to this, because its DI phase records every visited state whether or not a check flags it. Making ITERML robust therefore means relative species thresholds, sentinels (multi-step rollouts against DI) placed in every slow region and not only in the induction zone, and a gate that includes rollouts from the seeds; it also means that iterations must be allowed to fail, which is why the loop keeps the best network (iteration 1) as the parent of the next one.

Limitations.

- One laminar 2D configuration, one mechanism and one fuel. [TODO: second, turbulent case]
- Different parent networks (ood7 vs ood8) and one training seed per network (Section 4.1).
- Two ITERML iterations; the second was worse than the first and the loop's stop criteria have not been reached. A third iteration from iteration 1 with relative CO/OH checks, cooled-gas sentinels and two training seeds is planned. [TODO: iteration 3]
- The evaluation covers 0–50 ms of a transient; the steady state was tested with local time stepping for ISANN only (flame error 40–48 K

against a DI solution that itself drifts by 100 K, with CO frozen in cold long-residence gas).

- The circularity of ITERML: the seeds come from this burner, so the seeded data emphasise its states even though they are physics data; the a priori and ITERML networks are therefore reported side by side, and the offline benchmark measures the cost.

- No comparison with ISAT on the same machine.

7. Conclusions

Two ways of letting a reacting-flow simulation improve its chemistry surrogate were compared on a laminar methane/air burner with an 18-species mechanism, under one evaluation protocol in pure mode:

1. learning from the run's DI pairs (ISANN) gave the most accurate network: 2.1 K mean flame error, total CO within 3%, and first place on an offline benchmark of canonical states;
2. seeding physics-based exploration with the states where the network fails (ITERML) gave 3.3 K (1.7 K at 50 ms) after one iteration, with no CFD state in the training data, but too much CO and heat release in cooled lean gas and a small loss on the offline benchmark;
3. both schemes miss multi-step ignition errors with one-step checks; ISANN's network ignites the jet 0.4 ms early and ITERML's first and second networks 0.6 and 1.1 ms late;
4. the information the simulation gives about *where* the network fails is enough to reach a few-kelvin flame error without using the simulation's

results as labels.

A second ITERML iteration made the network worse (6.6 K, CO 6.4 times DI) because the one-step check that chooses the seeds is blind to slow, multi-step errors in cooled gas; in ITERML, the quality of the error detector, not the amount of seeded data, limits what the loop can learn, and an offline benchmark of canonical states does not replace a run in the target simulation.

Data and code availability

[TODO: Repository (chemmanifold-studio: openfoam/isann, openfoam/iterml, openfoam/protocol, scripts/19_iterml_seeds.py, chemmanifold/seeded.py), model files and protocol reports.]

Acknowledgements

[TODO:]

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